

scripts/scan-no-hardcoded-ipv4.sh

— User Guide

Last verified: 2026-05-15 (1.2.23 closure-cycle, §6.AD-debt task #61 backfill) **Inheritance:** HelixConstitution §11.4.18 (script documentation mandate)

Overview

Stub doc generated from the script's in-source header comment. See scripts/scan-no-hardcoded-ipv4.sh for canonical behavior. This stub exists so the CM-SCRIPT-DOCS-SYNC pre-push gate (.github/hooks/pre-push Check 9) starts gating modifications to this script.

The script's own header documentation (verbatim):

```
scripts/scan-no-hardcoded-ipv4.sh – standalone §6.R IPv4 scanner.
```

```
Purpose: enforce the §6.R No-Hardcoding Mandate clause that no
IPv4
literals appear in tracked source outside the exemption set.
§4.5.10 (CONTINUATION.md): the IPv4 enforcement is staged – this
scanner is the mechanical gate that lands the rule.
```

Exit codes:

- 0 – no IPv4 violations
- 1 – IPv4 violation(s) found (paths printed to stderr)

Exemptions (kept in lockstep with the §6.R clause body):

.env.example	– placeholder file
.lava-ci-evidence/	– forensic anchors +
matrix evidence	
docs/**/*.*md	– design docs,
plans, incident notes	
submodules/	– submodules
vendored at pinned hash	
*_test.go, *Test.kt, *Tests.kt, *Test.java	– synthetic test
fixtures	
src/test/, src/androidTest/	– test source roots
fixtures/	– test HTML/JSON
fixtures	
CHANGELOG.md	– release notes may
reference IPs in incident summaries	

`*.md, *.json, *.xml, *.yaml, *.yaml` — external config +
docs are legitimate home for
connection literals (Android
`network_security_config.xml` whitelists LAN
provisioning `.yaml` lists service ranges, Grafana
endpoints, etc.).
Code that READS these files is what §6.R
targets.
Loopback / docs-prefix IPs that are universally permitted (RFC
5737,
documentation-only) are filtered AFTER the file-level exemption so
a
code file that legitimately uses `127.0.0.1` or `0.0.0.0` still passes
—
those are not "hardcoded connection addresses" in the §6.R sense
(they cannot connect to anything but the local host or the
wildcard).

Usage

See the script's in-source comment block (above) for canonical usage examples.

Maintenance

When this script is modified, update this document in the same commit (CM-SCRIPT-DOCS-SYNC requires it). Per §11.4.18, the documentation MUST stay in sync with the codebase — no doc may be out of sync with its script.

Cross-references

- `scripts/scan-no-hardcoded-ipv4.sh` — the script itself
- `docs/helix-constitution-gates.md` — gate inventory
- `HelixConstitution Constitution.md` §11.4.18 (the mandate)
- `Lava CLAUDE.md` §6.AD (HelixConstitution Inheritance)